

Max-Relevance and Min-Redundancy Greedy Bayesian Network Learning on High Dimensional Data

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Abstract

Existing algorithms for learning Bayesian network require a lot of computation on high dimensional itemsets which affects accuracy especially on limited datasets and takes up a large amount of time. To address the above problem, we propose a novel Bayesian network learning algorithm MRMRG, Max Relevance-Min Redundancy Greedy. MRMRG algorithm is a variant of K2 which is a well-known BN learning algorithm. We also analyze the time complexity of MRMRG. The experimental results show that MRMRG algorithm has much better efficiency and accuracy than most of existing algorithms on limited datasets.

1. Introduction

There are many problems in fields as diverse as medical diagnosis, weather forecast, fault diagnosis, where there is a need for models that allow us to reason under uncertainty and take decisions, even when our knowledge is limited. To model this type of problems, AI community has proposed Bayesian network which allows us to reason under uncertainty. [3] In general, BN learning algorithms take one of two approaches: the constraint-based method and the search & score method. The constraint-based approach [8],[1] estimates from the data whether certain condition independences hold between variables. Typically, this estimation is performed using statistical or information theoretical measures. The search & score approach [2],[5] attempts to find a graph that maximizes the selected score. Score function is usually defined as a measure of fitness between the graph and the data. These algorithms use a score function in combination with a search method in order to measure the goodness of each explored structure from the space of feasible solutions. During the exploration process, the score function is applied in order to evaluate the fitness of each candidate structure to the data.

Although encouraging results have been reported, the two approaches both suffer some difficulties on limited data. With the constraint-based approach, a statistical or information theoretical measure may become unreliable on limited datasets. If the measure would return incorrect independence statements, errors could arise in graphical structure. The search & score approach suffers from the exponential search space. The search space is so vast that the category algorithms have to use heuristic methods to find approximately optimal Bayesian network.

The K2 algorithm [2] is a typical search & score method. Although it has already been presented for 15 years, K2 is still one of the most effective BN learning algorithms. Moreover, K2 is often used by other BN learning algorithms in order to improve performance. In this paper, we propose MRMRG algorithm that improve K2 for limited datasets. MRMRG algorithm incorporates Max Relevance - Min Redundancy feature selection technology to obtain better efficiency and accuracy when learning on limited datasets. MRMRG algorithm also proposes Local Bayesian Increment function to avoid overfitting and improve accuracy.

2. Concepts and theorems

2.1. Bayesian network

A Bayesian network is defined as a pair $B = \{G, \Theta\}$, where G is a directed acyclic graph $G = \{V(G), A(G)\}$, with a set of nodes $V(G) = \{V_1, \dots, V_n\}$, representing a set of stochastic variables and a set of arcs $A(G) \subseteq V(G) \times V(G)$, representing independent relationships that exist between variables. Θ represents the set of parameters that quantifies the network. It contains a parameter $\theta = P(x_i | \phi_i[j])$ for each possible value x_i of X_i , and $\phi_i[j]$ of π_i . Here π_i denotes the set of parents of X_i in G and $\phi_i[j]$ is a particular instantiation of the parents set π_i .

2.2. MRMR and mutual information

In feature selection, Maximum Mutual Information (MMI) [4] is the method which is to find a feature set S_m with m features $\{x_i\}$, which jointly have the largest mutual information on the target class c , $x_m = \arg \max_{S_m} I(S_m; c)$.

In feature selection, Max Relevance-Min Redundancy (MRMR) [4] is that suppose we already have the feature set with $m-1$ features S_{m-1} , if we want to select one feature from the set $\{X - S_{m-1}\}$, we can select the feature x_m that maximizes the following formula:

$$x_m = \arg \max_{x_j \in X - S_{m-1}} \left[I(x_j; c) - \frac{1}{m-1} \sum_{x_i \in S_{m-1}} I(x_j; x_i) \right] \quad (1)$$

Lemma 2. [MRMR and Maximum Mutual Information]

If one feature is added at one time, MRMR is equivalent to MMI.

Note that proof of Lemma 2 refer to Section 2.3 in [4].

3. K2 algorithm

Given a complete dataset D , K2 searches for the Bayesian network G^* with maximal $P(G, D)$.

Let D be a dataset of m cases, where each case contains a value for each variable in V . D is sufficiently large. Let V be a set of n discrete variables, where x_i in V has r_i possible values $(v_{i1}, v_{i2}, \dots, v_{ir_i})$. Let G denote a Bayesian network structure containing just the variables in V . Each variable x_i in G has the parents set π_i . Let $\phi_i[j]$ denote the j^{th} unique instantiation of π_i relative to D . Suppose there are q_i such unique instantiation of π_i . Define N_{ijk} to be the number of cases in D in which variable x_i is instantiated as v_{ik} and π_i is instantiated as $\phi_i[j]$. Let $N_{ij} = \sum_{k=1}^{r_i} N_{ijk}$.

Given a Bayesian network model, cases occur independently. Bayesian network prior distribution is uniform. It follows $g(i, \pi_i) = \prod_{j=1}^{q_i} \frac{(r_i-1)!}{(N_{ij}+r_i-1)!} \prod_{k=1}^{r_i} N_{ijk}!$.

It starts by assuming that a node has no parents, and then in every step it adds incrementally the node which can most increase the probability of the resulting BN, to the parents set. K2 stops adding nodes to parents set when the addition cannot increase the probability of the BN given the data.

K2 algorithm

Input: A set of n nodes $V = \{x_1, x_2, \dots, x_n\}$, an ordering on the nodes, an upper bound u_{max} on the number of parents a node may have, Pre_i denotes the set of nodes that precede x_i , and a dataset $D = \{d_1, \dots, d_m\}$ containing m cases. The dataset D is sufficiently large.

Output: For each node x_i , a printout of the parents set π_i of the node.

Procedure:

```

for i=1 to n do
   $\pi_i = \text{NULL};$ 
   $P_{old} = g(i, \pi_i);$ 
  OK=TRUE;
  while OK and ( $|\pi_i| < u_{max}$ ) do
     $y = \arg \max_{x_j \in Pre_i - \pi_i} [g(i, \pi_i \cup x_j)];$ 
     $P_{new} = g(i, \pi_i \cup x_j);$ 
    if  $P_{new} > P_{old}$  then
       $P_{old} = P_{new};$ 
       $\pi_i = \pi_i \cup \{y\};$ 
    else
      OK=FALSE;
    end if
  end while
  Output( $x_i, \pi_i$ );
end for

```

4. Local bayesian increment function

Let X and Y be two discrete variables, Z be a set of discrete variables, and z be an instantiation for Z . $X, Y \notin Z$.

According to Moore's recommendation [7] about chi-squared test, we assume that the dataset D satisfying at least one of the following two conditions is "sufficiently large" for $\{XUY\}$:

1. The number of cases of the dataset D is much larger than the number of the values of $\{XUY\}$, such as $\|D\| > 5 \times (\|X\| \times \|Y\|)$;
2. All cells of $\{XUY\}$ in the contingency table have expected value greater than 1, and at least 80% of the cells in the contingency table about $\{XUY\}$ have expected value greater than 5.

According to Moore's recommendation [7] about chi-squared test, we assume that the sub-dataset $D_{Z=z}$ satisfying at least one of the following two conditions is "locally sufficiently large" for $\{XUY\}$ given $Z=z$:

1. The number of cases of the sub-dataset $D_{Z=z}$ is much larger than the number of the values of $\{XUY\}$, such as $\|D_{Z=z}\| > 5 \times (\|X\| \times \|Y\|)$;
2. All cells of $\{XUY\}$ in the contingency table conditioned on $Z=z$ have expected value greater than 1, and at least 80% of the cells in the contingency table about $\{XUY\}$ on $Z=z$ have expected value greater than 5.

Let D be a dataset of m cases, where each case contains a value for each variable. Let V be a set of n

discrete variables, where x_i in V has r_i possible values ($v_{i1}, v_{i2}, \dots, v_{ir_i}$). B_P and B_S denote Bayesian network structures containing just the variables in V . B_S exactly has one edge $y \rightarrow x_i$ more than B_P . Variable x_i in B_P has the parents set π_i . Variable x_i in B_S has the parents set $\pi_i \cup y$. Let $\phi_i[j]$ denote the j^{th} particular instantiation of π_i relative to D . Suppose there are q_i such unique instantiations of π_i . Define N_{ijk} to be the number of cases in D in which variable x_i is instantiated as v_{ijk} and π_i is instantiated as $\phi_i[j]$.

$$\text{Let } N_{ijk} = \sum_y N_{i, \{y \cup y\}, k}, N_{ij} = \sum_{k=1}^{r_i} N_{ijk}.$$

$\theta_{ijk}^{B_P}$ denotes $P(x_i = v_{ijk} \mid \pi_i = \phi_i[j])$, $\theta_{ijk}^{B_P} > 0$, $\sum_{k=1}^{r_i} \theta_{ijk}^{B_P} = 1$, $\theta_{ij}^{B_P} = \cup_{j=1}^{q_i} \{\theta_{ijk}^{B_P}\}$, $\theta_i^{B_P} = \cup_{j=1}^{q_i} \{\theta_{ij}^{B_P}\}$, $\theta^{B_P} = \cup_{i=1}^n \{\theta_i^{B_P}\}$. $\hat{\theta}_{ijk}^{B_P}, \hat{\theta}_{ij}^{B_P}, \hat{\theta}_i^{B_P}, \hat{\theta}^{B_P}$ denote the maximum likelihoods of $\theta_{ijk}^{B_P}, \theta_{ij}^{B_P}, \theta_i^{B_P}, \theta^{B_P}$.

Given a Bayesian network model, cases occur independently. Bayesian network prior distribution is uniform. Given a Bayesian network model, there exist two properties: Parameter Independence and Parameter Modularity. [5]

In order to control the complexity of BN model, we apply the BIC (Bayesian Information Criterion) approximation formula[6]

$$\begin{aligned} BIC(B_S) &= \left[\log(L(\hat{\Theta})) - \frac{1}{2} \log(m) \dim(\hat{\Theta}) \right] \\ &\approx \log(P(D \mid B_S)) \end{aligned}$$

to $\log(P(B_S, D)/P(B_P, D))$. BIC adds the penalty of structure complexity to Local Bayesian Increment function in order to avoid overfitting.

On limited datasets, with the dimension increase of current parents set π_i , the maximum likelihood estimations $\hat{\Theta}$ of parameters Θ in Bayesian network will be more and more unreliable because the dataset D does not satisfy either condition of "sufficiently large" for π_i . But we find that there are some specific instantiations of π_i that the sub-datasets $D_{\pi_i=\phi_i[j]}$ are "locally sufficiently large" for $\{X \cup Y\}$ given $\pi_i = \phi_i[j]$.

Definition 1. [Local Bayesian Increment Function]

$$\begin{aligned} Lbi(y, i, \pi_i) &= \log(P(B_S, D)/P(B_P, D)) \\ &\approx BIC(B_S) - BIC(B_P) \\ &= \log(L(\hat{\Theta}^{B_S})/L(\hat{\Theta}^{B_P})) \\ &\quad - \frac{1}{2} \log(m) [\dim(\hat{\Theta}^{B_S}) - \dim(\hat{\Theta}^{B_P})] \end{aligned} \quad (2)$$

$$\begin{aligned} &\log(L(\hat{\Theta}^{B_S})/L(\hat{\Theta}^{B_P})) \\ &= \log(P(D \mid \hat{\Theta}^{B_S})) - \log(P(D \mid \hat{\Theta}^{B_P})) \end{aligned}$$

$$\begin{aligned} &= \sum_{l=1}^m \left(\log(P(d_l \mid \hat{\Theta}^{B_S})) - \log(P(d_l \mid \hat{\Theta}^{B_P})) \right) \\ &= \sum_{l=1}^m \log(P(x_i^l \mid \hat{\Theta}_i^{B_S}, \pi_i^l \cup y) / P(x_i^l \mid \hat{\Theta}_i^{B_P}, \pi_i^l)) \end{aligned} \quad (3)$$

According to the definition of overfitting(the likelihood of overfitting for the training dataset is no less than non-overfitting), we assume that the log-likelihood does not change on the sub-datasets $D_{\pi_i=\phi_i[*]}$ which are not "locally sufficiently large" for $\{X \cup Y\}$ (that is, to assume that there is overfitting for X and the set of its parents),

$$\begin{aligned} &\sum_{d_l \in D_{\pi_i=\phi_i[*]}} \log P(x^l \mid \hat{\Theta}^{B_S}, \pi_l \cup y) \\ &= \sum_{d_l \in D_{\pi_i=\phi_i[*]}} \log P(x^l \mid \hat{\Theta}^{B_P}, \pi_l). \end{aligned} \quad (4)$$

According to (3) and (4), we infer the following results:

$$\begin{aligned} &\log(L(\hat{\Theta}^{B_S})) - \log(L(\hat{\Theta}^{B_P})) \\ &= \sum_j \sum_{k=1}^{r_i} \sum_y N_{i, \{y \cup y\}, k} \log \left(\frac{N_{i, \{y \cup y\}, k} N_{ij}}{N_{i, \{y \cup y\}} N_{ijk}} \right), \\ &\quad \text{for } j, D_{\pi_i=\phi_i[j]} \text{ is "locally sufficiently large"} \\ &= \sum_j N_{ij} \times I_j(X, Y), \\ &\quad \text{for } j, D_{\pi_i=\phi_i[j]} \text{ is "locally sufficiently large"} \end{aligned} \quad (5)$$

$$\dim(\hat{\Theta}^{B_S}) - \dim(\hat{\Theta}^{B_P}) = (r_y - 1)(r_i - 1)q_i \quad (6)$$

$$\begin{aligned} &Lbi(y, i, \pi_i) \\ &= \sum_j N_{ij} \times I_j(X, Y) - \frac{1}{2} (r_y - 1)(r_i - 1)q_i \log(m), \\ &\quad \text{for } j, D_{\pi_i=\phi_i[j]} \text{ is "locally sufficiently large"}. \end{aligned} \quad (7)$$

Note: $I_j(X, Y)$ is the mutual information between variable X and variable Y on the sub-dataset $D_{\pi_i=\phi_i[j]}$.

5. MRMRG algorithm

In this section, we discuss MRMRG algorithm, Max Relevance-Min Redundancy Greedy algorithm. On limited datasets, MRMRG algorithm improves K2 in two ways.

Firstly, on limited dataset, the computation of $g(i, \pi_i)$ has less and less reliability and robustness with the dimension increase of π_i , so that the formula $y = \arg \max_{x_j \in Pre_i - \pi_i} [g(i, \pi_i \cup x_j)]$ cannot accurately obtain

node y which makes mutual information increment between nodes x_i and $\pi_i \cup y$ maximal in K2 algorithm. According to Lemma 2, MRMRG algorithm replaces $y = \arg \max_{x_j \in Pre_i - \pi_i} [g(i, \pi_i \cup x_j)]$ by

$$y = \arg \max_{x_j \in Pre_i - \pi_i} \left[I(x_i, x_j) - \frac{1}{|\pi_i|} \sum_{x \in \pi_i} I(x, x_j) \right].$$

MRMRG reduces the dimension of the computation by 2 dimension. Therefore, MRMRG algorithm improves the computational reliability and robustness of previous BN learning algorithms on limited datasets.

Secondly, When the dataset D is “sufficiently large” for $\{X \cup Y \cup \pi_i\}$, MRMRG algorithm uses Local Bayesian Increment function to control the complexity of BN and avoid overfitting. When dataset D is not “sufficiently large” for $\{X \cup Y \cup \pi_i\}$, but there exist sub-datasets $D_{\pi_i = \phi_i[j]}$ that are “locally sufficiently large” for $\{X \cup Y\}$ given $\pi_i = \phi_i[j]$, MRMRG algorithm can also apply Local Bayesian Increment function to improve accuracy and avoid overfitting. The technique also makes it unnecessary to set the maximal parents number u_{max} of a node.

5.1. MRMRG algorithm

Input: A set of n nodes, an ordering on the nodes, the mutual information between any two nodes is saved in 2-dimension array $MI(n, n)$, Pre_i denotes the set of nodes that precede x_i , π_i denotes the current parents set of node x_i , a dataset containing m cases.

Output: For each node x_i , a printout of the parents set of the node.

Assume that all conditions and assumptions in section 4 are satisfied.

Procedure:

```

/* get mutual information between any two nodes */
initialize  $MI(n, n)$  to 0;
for  $i=1$  to  $n$  do
    for  $j=i+1$  to  $n$  do
         $MI[i, j] = MI[j, i] = I(x_i, x_j)$ ;
    end for
end for
/* obtain a set of parents of each node */
for  $i=1$  to  $n$  do
     $\pi_i = \text{NULL}$ ;
    OK=TRUE;
    while OK do
         $y = \arg \max_{x_j \in (Pre_i - \pi_i)} \left[ I(x_i, x_j) - \frac{1}{|\pi_i|} \sum_{x \in \pi_i} I(x, x_j) \right]$ ;
        if  $Lbi(y, i, \pi_i) > 0$  then
             $\pi_i = \pi_i \cup \{y\}$ ;
        else
            OK=FALSE;
        end if
    end while
end for

```

```

end if
end while
Output( $x_i, \pi_i$ );
end for

```

MRMRG algorithm includes two parts:

The first part is the computation of mutual information between any two nodes.

In the second part, for the node x_i , MRMRG algorithm initializes the current parents set π_i of given node x_i as NULL at first, and then add the nodes which can get the maximal increment of mutual information into the current parents set π_i one by one, until the computation result of Local Bayesian Increment function is no more than 0. Repeating the above steps for every node, we can obtain an approximately optimal Bayesian network.

5.2. Time complexity of MRMRG

The complexity of the first part of MRMRG is $O(n^2)$.

In the second part of MRMRG, the complexity of

$$y = \arg \max_{x_j \in Pre_i - \pi_i} \left[I(x_i, x_j) - \frac{1}{|\pi_i|} \sum_{x \in \pi_i} I(x, x_j) \right]$$

is $O(n)$. The complexity of $Lbi(y, i, \pi_i)$ is $O(mnr)$ (Note: Let $r = \max(r_i) i = 1, \dots, n$). The other steps in the while statement require $O(1)$ time. The while statement loops at most $O(n)$ times, each time it is entered. The for statement loops n times. In the worst case, the complexity of second part in MRMRG is $(O(n) + O(mnr)) * O(n) * n = O(mn^3r)$.

The overall time complexity of MRMRG is $O(n^2) + O(mn^3r) = O(mn^3r)$ at worst. On the other hand, the worst-case time complexity of K2 is $O(mn^4r)$.

6. Experimental results

We implemented MRMRG algorithm, K2 algorithm, TPDA algorithm and presented the comparison of the experimental results for 3 implementations.

Tests were run on a PC with Pentium4 1.5GHz and 1GB RAM. The operating system was Windows 2000. The program was developed under Matlab 7.0. 3 Bayesian networks were used. Table 1 shows the characteristics of these networks. The characteristics include the number of nodes, the number of arcs, the max number of node parents/children(Max In/Out-Degree), and the min/max number of node values(Domain Range). From the networks, we randomly sampled 3 datasets with 500, 2000, 10000 training cases each.

6.1. Comparison of runtime among algorithms

A summary of the time results of the execution of all three algorithms is in Table 2. We normalized the times reported by dividing by the corresponding running time of MRMRG on the same datasets and reported the averages over sample sizes. Thus, a normalized running time of greater than 1 implies a slower algorithm than MRMRG on the same learning task. A normalized running time of lower than 1 implies a faster algorithm than MRMRG.

From the results, we can see that MRMRG has better efficiency than other two algorithms K2 and TPDA. For small sample sizes (500, 2000), MRMRG runs several times faster than K2 and TPDA. For larger sample sizes (10000), MRMRG performs nearly one magnitude faster than K2.

6.2. Comparison of accuracy among algorithms

We compared the accuracy of Bayesian networks learned by these 3 algorithms according to the BDeu score. The BDeu score corresponds to the posteriori probability of the structure learned.[5] Table 3-5 reports the results.

From the results, we can see that MRMRG can learn more accurately than K2 and TPDA on limited datasets, such as Insur(500,2000), Alarm(500,2000), Munin(500,2000,10000). For almost sufficiently large datasets, the accuracy of MRMRG and K2 is almost same, such as Insur(10000), Alarm(10000).

7. Conclusions

Efficiency and accuracy are two main indices in evaluating algorithms for learning Bayesian network. MRMRG algorithm greatly reduces the number of high dimensional computations and improves scalability of learning. The experimental results indicate that MRMRG has better performance on efficiency and accuracy than most of existing algorithms on limited datasets.

We are interesting in incorporate ordering-based search method into our MRMRG algorithm to implement MRMRG without the information of the order between nodes. In addition, our results are preliminary and much more experimentation is needed on different network structures.

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Table 1. Bayesian networks

BN	Vars Num	Arcs Num	Max In/Out- Degree	Domain Range
Insur	27	52	3/7	2-5
Alarm	37	46	4/5	2-4
Munin	189	282	3/15	1-21

Table 2. Normalized Runtime of Algorithms

Size	MRMRG	K2	TPDA
500	1.0	1.58	3.79
2000	1.0	2.39	3.26
10000	1.0	8.98	2.18

Table 3. Bayesian Score BDeu(Insur)

Size	MRMRG	K2	TPDA
500	-19.0322	-19.5833	-28.7857
2000	-18.0233	-18.2722	-24.7111
10000	-17.8487	-17.8498	-17.8571

Table 4. Bayesian Score BDeu(Alarm)

Size	MRMRG	K2	TPDA
500	-13.8577	-13.9500	-18.0973
2000	-13.1732	-13.3824	-14.5286
10000	-12.9801	-12.9644	-12.9640

Table 5. Bayesian Score BDeu(Munin)

Size	MRMRG	K2	TPDA
500	-62.4827	-63.2500	-125.6250
2000	-60.1359	-61.3175	-143.4762
10000	-59.1268	-59.8576	-110.3646

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